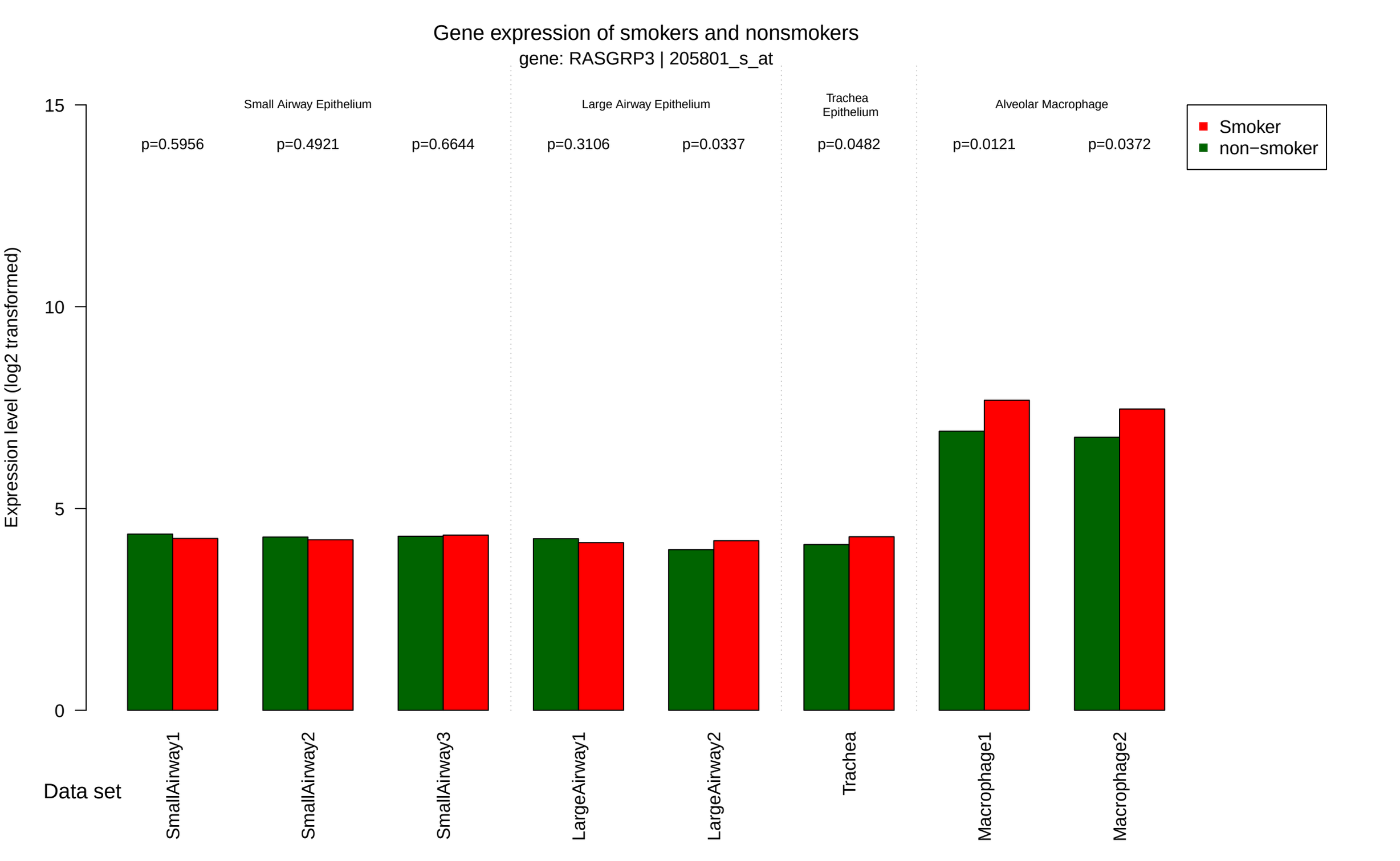
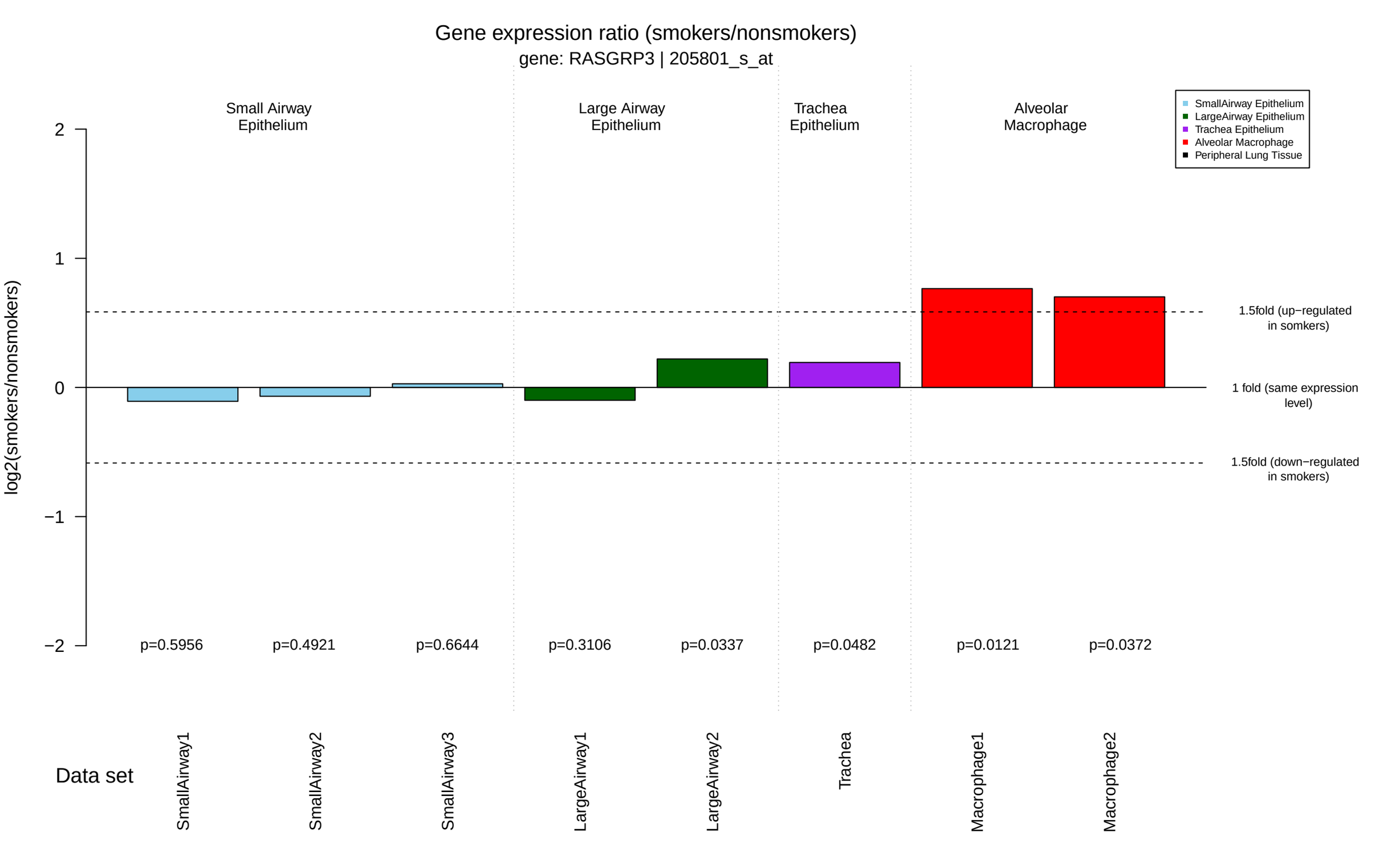
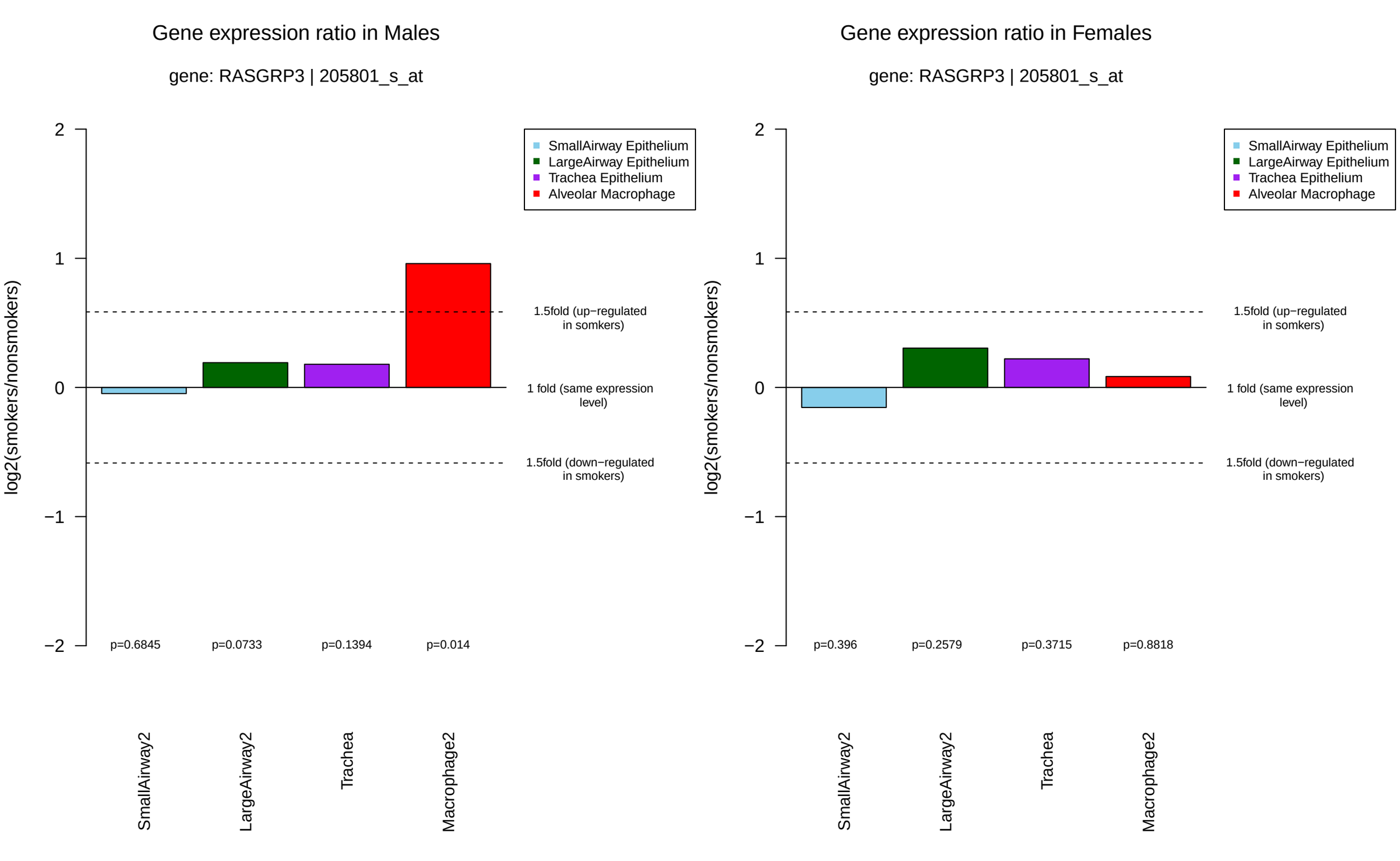
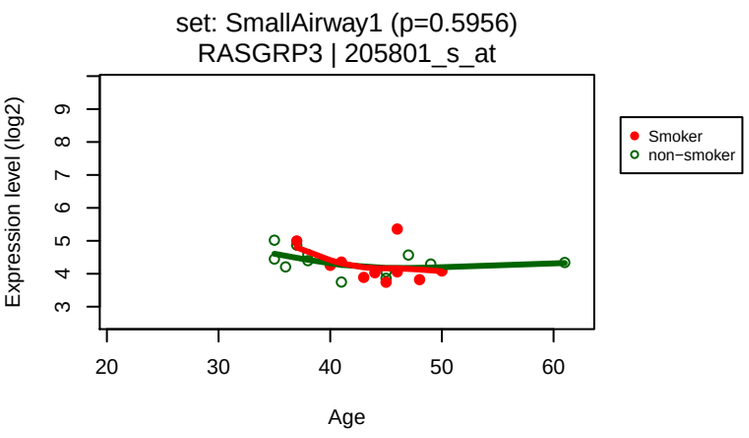
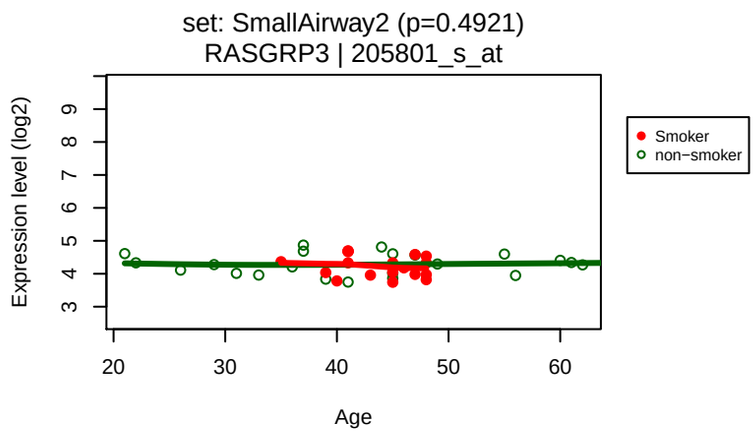
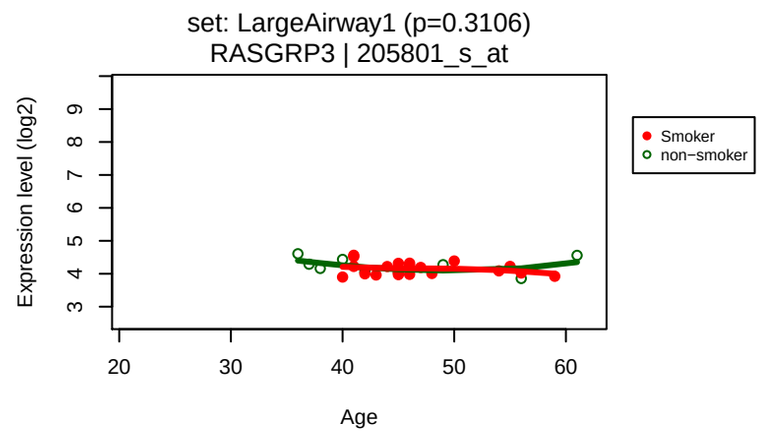

Gene expression of smokers and nonsmokers
gene: RASGRP3 | 205801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5956
p=0.4921
p=0.6644
p=0.3106
p=0.0337
p=0.0482
p=0.0121
p=0.0372
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

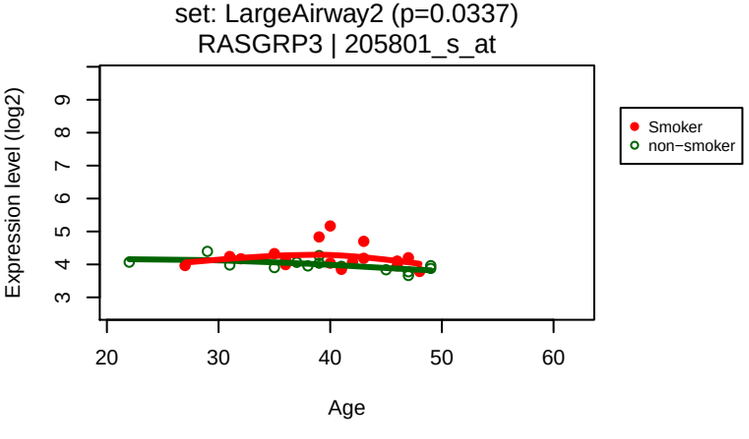

Gene expression ratio (smokers/nonsmokers)
gene: RASGRP3 | 205801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5956
p=0.4921
p=0.6644
p=0.3106
p=0.0337
p=0.0482
p=0.0121
p=0.0372
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

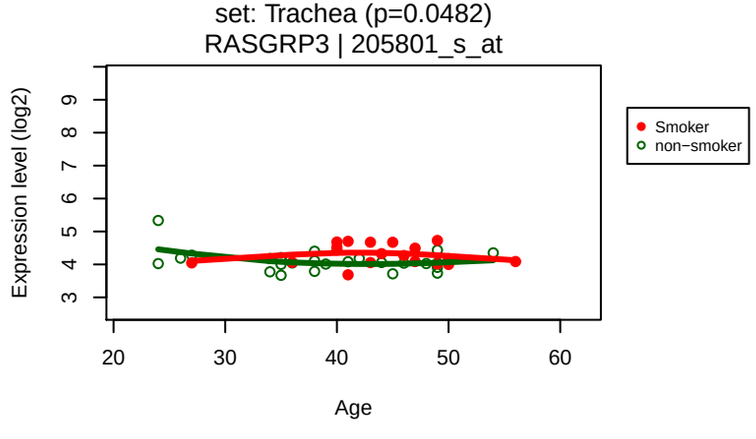

Gene expression ratio in Males
gene: RASGRP3 | 205801_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6845
p=0.0733
p=0.1394
p=0.014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRP3 | 205801_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.396
p=0.2579
p=0.3715
p=0.8818
SmallAirway2
LargeAirway2
Trachea
Macrophage2

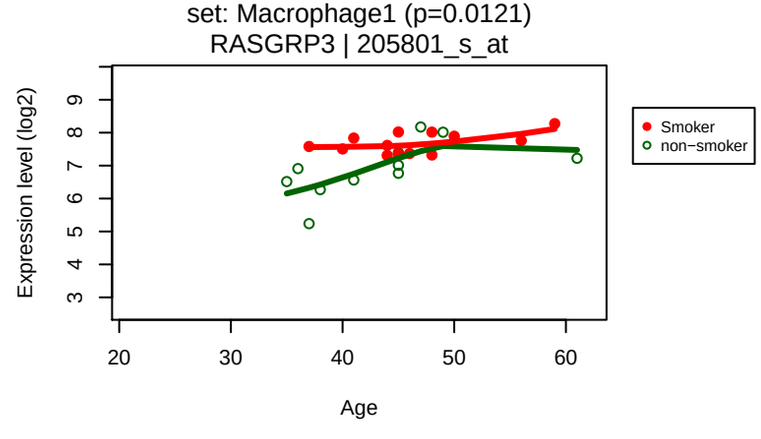

set: SmallAirway1 (p=0.5956)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker

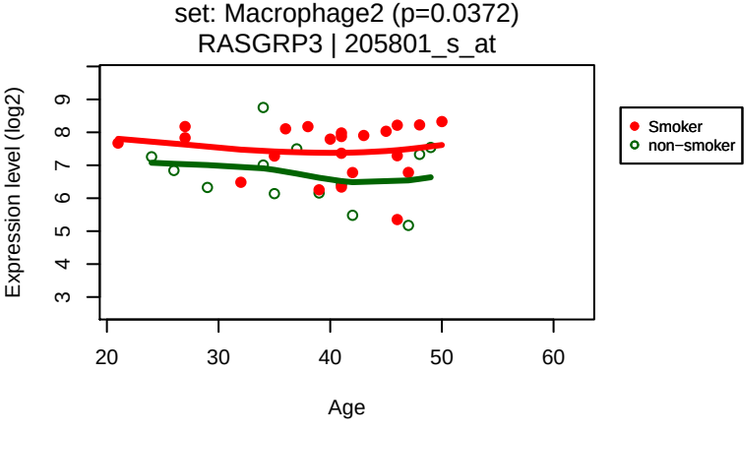

set: SmallAirway2 (p=0.4921)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker

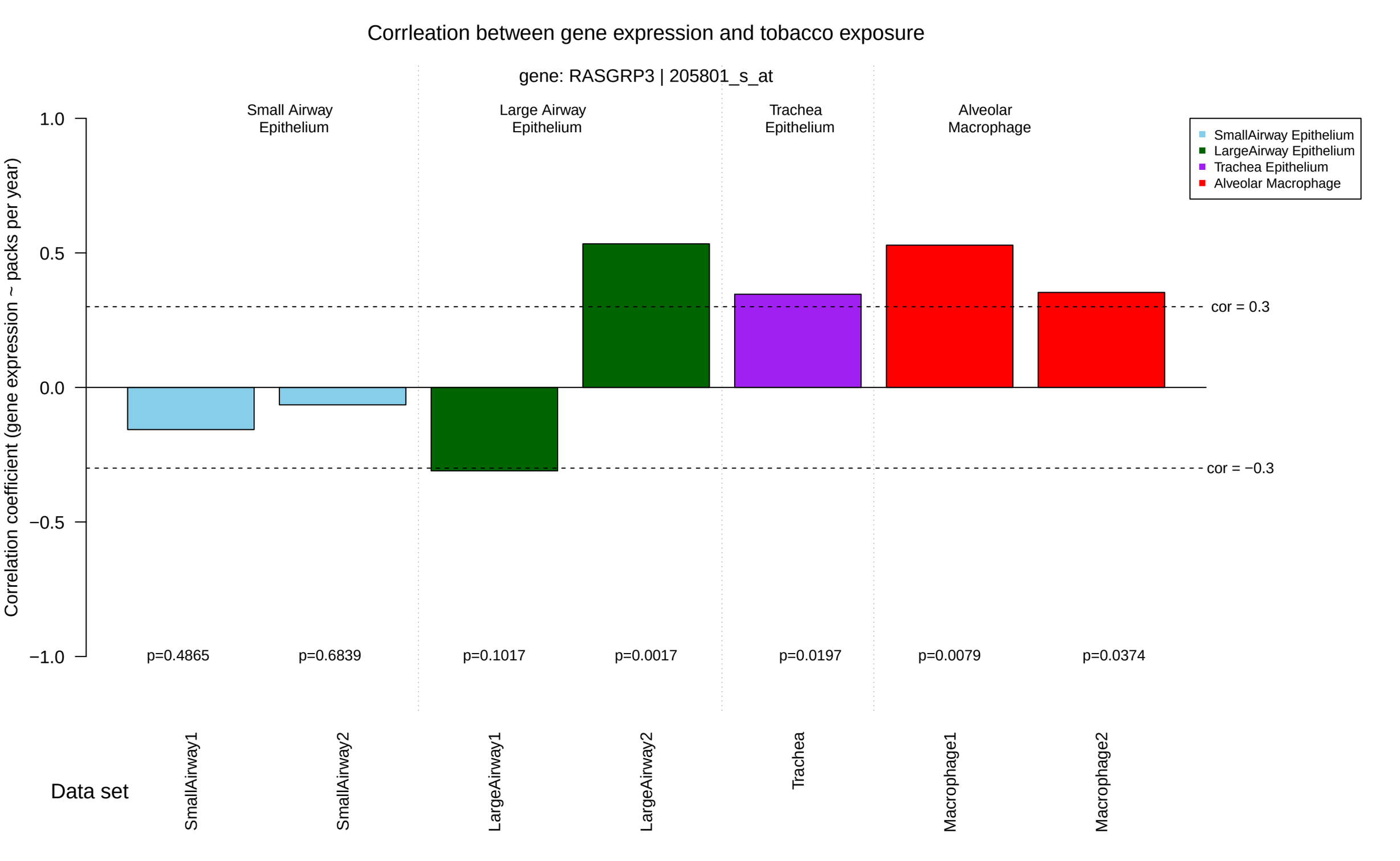

set: LargeAirway1 (p=0.3106)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0337)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0482)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0121)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0372)
RASGRP3 | 205801_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRP3 | 205801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4865
p=0.6839
p=0.1017
p=0.0017
p=0.0197
p=0.0079
p=0.0374
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2